Notes: Chen, Ren, and Zha (AER, 2018)

The Nexus of Monetary Policy and Shadow Banking in China

Contents

1 Big picture

- **Question.** How does monetary policy transmit to credit in China when (i) the PBoC targets money growth (M2) and (ii) banks can shift credit creation into *shadow banking*?
- **Core finding.** When exogenous M2 growth falls (a tightening shock), *traditional bank loans contract* but *shadow-banking credit expands*—especially at **nonstate** banks. The expansion partly offsets the tightening so that **total bank credit does not fall much** (and can even rise in the short run).
- **Mechanism (intuition).** Tightening reduces *deposit funding* (bank liabilities). With loan-to-deposit and other regulations, banks facing tighter constraints (nonstate banks) have incentives to *re-route credit* into categories that evade constraints: (i) facilitating more off-balance-sheet entrusted loans and (ii) bringing shadow-banking assets onto their balance sheets (ARIX).
- **Why this matters.** It provides a concrete reason why monetary policy may have weaker (or different) credit effects in economies with large and adaptable shadow-banking sectors.

2 Institutional setting and objects

2.1 Two shadow-banking objects in the paper

- **Off-balance-sheet: Entrusted loans.** A nonfinancial lender makes a loan to a risky borrower *through a trustee* (often a bank). The trustee facilitates/records the transaction but the loan is not a standard bank loan.
- **On-balance-sheet: ARIX (Accounts Receivable Investment).** A balance-sheet category used by banks to hold investment assets *outside traditional loans*. The paper treats ARIX as capturing on-balance-sheet shadow-banking assets (e.g., purchases of trust/entrusted-loan beneficiary rights).

2.2 Figure 5 (diagram): the "nexus"

Interpretation of Fig. 5 (conceptual map).

- Monetary policy works through the banking system primarily via deposits and reserves (and open market operations through primary dealers).
- When deposits/reserves become tight, regulated *bank loans* are constrained.
- Banks can respond by (i) facilitating off-balance-sheet entrusted lending and (ii) holding shadow-banking exposures on the balance sheet (ARIX), both of which can bypass some constraints that bind traditional loans.

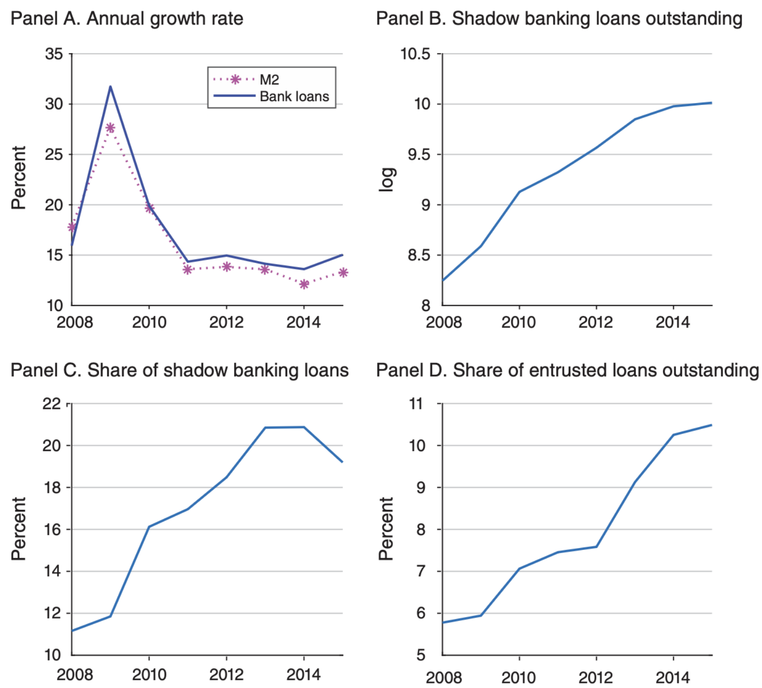

FIGURE 1. AGGREGATE TIME SERIES OF M2, BANK LOANS, AND SHADOW BANKING VARIABLES

Notes: The shadow banking loan volume is the sum of entrusted lending, trusted lending, and bank acceptances, all of which are off balance sheet. The share of shadow banking loans is the ratio of shadow banking loans to the sum of shadow banking loans and bank loans. The share of entrusted loans is the ratio of entrusted loans to the sum of entrusted loans and bank loans.

3 Stylized facts about credit and shadow banking

3.1 Figure 1: bank loans vs. shadow-banking loans over 2009–2015

What Fig. 1 shows (my reading).

- **Panel A:** M2 growth and bank-loan growth fall after the stimulus period. (China tightens money growth post-2009.)
- **Panel B:** the *level* of shadow-banking loans rises strongly (log outstanding increases).
- **Panel C:** shadow-banking loans become a larger share of total bank credit.
- **Panel D:** the share of new entrusted loans in new credit rises.

Intuition: even when money growth slows, shadow-banking activity continues to expand, hinting at substitution away from traditional loans.

3.2 Table 1: bank loans vs. entrusted loans (maturity and rates)

Table 1 takeaway.

- Entrusted loans have **shorter maturity** (about 21 months) than bank loans (about 31 months).

- Entrusted loans have **higher interest rates** (about 7.6%) than bank loans (about 6.5%).

Interpretation: entrusted loans look like riskier/less-regulated credit: shorter duration, higher pricing.

Table 1—Bank Loans versus Entrusted Loans (*Average for 2009–2015*)

Loan type	Loan maturity	Loan rate
Bank loans	30.91	6.51
Entrusted loans	20.99	7.59

Notes: Loan maturity is expressed in months and loan rate in percent. Both measures are averages weighted by loan amount.

Source: CEIC and our constructed entrusted loan dataset

Table 2—Capital Adequacy Ratios (CAR), Excess Reserve Ratio (ERR), and LDR for State and Nonstate Banks in 2009–2015 (*Percent*)

Description	CAR	ERR	LDR (EOY)	LDR (average)
State banks	13.07	1.45	68.06	68.69
Nonstate banks	12.16	3.32	73.02	76.68
Overall	12.71	1.90	69.15	70.4
SE	4.49	0.46	1.48	1.86
p-value	0.85	0.00	0.4	0.27

Notes: EOY stands for end of year. Average means an average value of the LDRs within each year for each bank. Each ratio for each year is weighted by bank assets and the reported ratio is a simple average across years. The calculation is based on the balance-sheet information of commercial banks reported by Bankscope and WIND. Capital adequacy ratios for all commercial banks are available from Bankscope. LDRs and excess reserve ratios are manually collected from annual reports of publicly listed banks, which are available from WIND. We compute the excess reserve ratio of each bank for every year, calculate a weighted average of these ratios for all the banks within each group (the state group and the nonstate group) for each year, and then take a simple average of these ratios across years. "SE" stands for standard error. The standard error for the difference between the ratio for state banks and the ratio for nonstate banks, along with the corresponding *p*-value (round to two decimal places), is reported in the last two rows of the table.

3.3 Table 2: bank constraints (CAR, excess reserves, LDR) by bank type

Table 2 takeaway.

- **Capital adequacy ratio (CAR):** state and nonstate banks look similar on average and comfortably above the 8% minimum.
- **Excess reserve ratio (ERR):** nonstate banks hold *more* excess reserves on average (statistically significant difference), suggesting more caution or stronger liquidity management.
- **Loan-to-deposit ratio (LDR):** nonstate banks have higher average LDRs (closer to the 75% regulatory ceiling historically used in China), though the mean differences are not always statistically strong in this table.

Interpretation: nonstate banks look more "constrained at the margin" in liquidity/funding terms, a key place where monetary tightening can bite.

4 Measuring Chinese monetary policy: an endogenous policy rule + exogenous shocks

4.1 The monetary policy rule (Eq. (1))

The authors model the PBoC as targeting M2 growth with an *endogenously switching* reaction to output shortfalls:

$$g_{m,t} = \gamma_0 + \gamma_m g_{m,t-1} + \gamma_\pi\big(\pi_{t-1} - \pi^*\big) + \gamma_{x,t}\big(g_{x,t-1} - g^*_{x,t-1}\big) + e_{m,t}, \tag{1}$$

$$\gamma_{x,t} = \begin{cases} \gamma_{x,a}, & \text{if } g_{x,t-1} - g^*_{x,t-1} \geq 0, \\ \gamma_{x,b}, & \text{if } g_{x,t-1} - g^*_{x,t-1} < 0, \end{cases} \tag{2}$$

$$e_{m,t} = \sigma_{m,t}\,\varepsilon_{m,t}, \qquad \varepsilon_{m,t} \sim \mathcal{N}(0,1), \tag{3}$$

$$\sigma_{m,t} = \begin{cases} \sigma_{m,a}, & \text{if } g_{x,t-1} - g^*_{x,t-1} \geq 0, \\ \sigma_{m,b}, & \text{if } g_{x,t-1} - g^*_{x,t-1} < 0. \end{cases} \tag{4}$$

Definitions.

- $g_{m,t}$: quarterly M2 growth.
- π_t: CPI inflation; π^* is the inflation target.
- $g_{x,t}$: real GDP growth; $g^*_{x,t}$ is the GDP-growth target.
- The key state variable is the *GDP growth shortfall* $g_{x,t-1} - g^*_{x,t-1}$.

4.2 Figure 2: endogenous vs. exogenous M2 growth

What Fig. 2 shows.

- Panels A–B: GDP growth shortfall and inflation relative to targets (policy objectives).
- Panel C: actual M2 growth vs. the *endogenous* (systematic) component from (1); they track closely.
- Panel D: the *exogenous* component (policy shocks). This is the series later used to study causal effects of monetary policy on shadow banking.

Interpretation: most variation in M2 growth is systematic, but the residual (shock) component is sizable and is used as an instrument for monetary policy innovations.

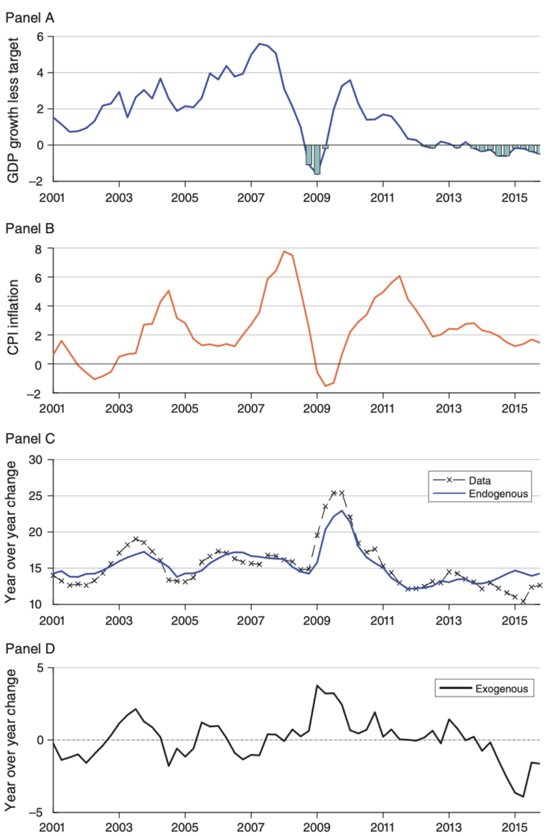

Figure 2. Ultimate Goals of Monetary Policy and Its Endogenous and Exogenous Components

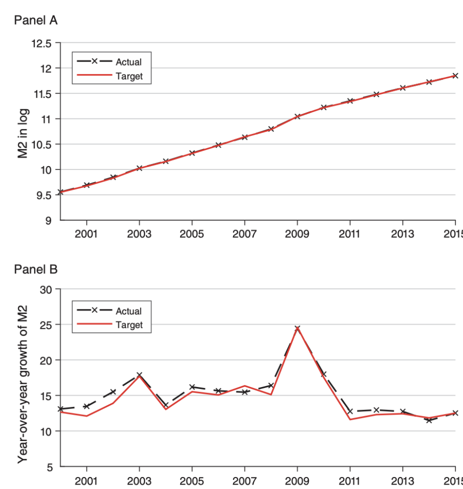

Figure 3. Actual and Targeted End-of-Year M2 Series

Notes: The government sets the M2 growth target each year as year-over-year M2 growth at the end of the year. Note that the end-of-year M2 growth (year-over-year) rates are slightly different from the annual M2 growth rates based on the average M2 stock for each year (panel A of Figure 1). Year-over-year growth rates are expressed in percent.

Source: Various RWG issues published by the State Council and PBC

4.3 Figure 3: actual vs. targeted M2

Fig. 3 shows that actual end-of-year M2 (levels and growth rates) is close to the PBoC target path. **Interpretation:** it is reasonable to view M2 growth as a policy instrument/target in this setting.

4.4 Figure 4: reserve requirement dynamics

Fig. 4 plots reserve requirement ratios and (excess) reserve ratios. **Interpretation:** reserve requirements are a complementary policy tool, and fluctuations in excess reserves indicate varying liquidity conditions for banks.

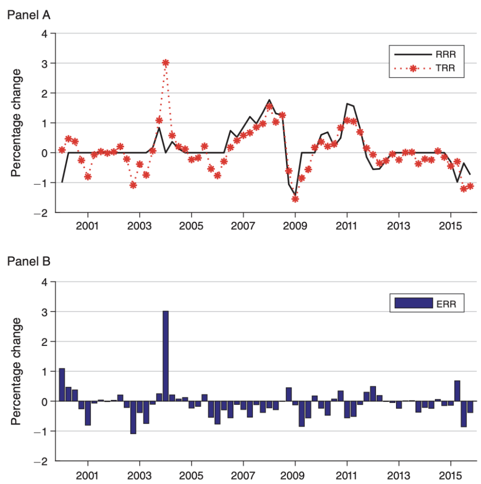

Figure 4. Quarterly Time Series of the Reserve Requirement Ratio (RRR), Total Reserve Ratio (TRR), and Excess Reserve Ratio (ERR)

Source: PBC and CEIC

Table 3—Estimated Monetary Policy

Coefficient	Estimate	SE	p-value
γ_m	0.391	0.101	0.000
γ_π	−0.397	0.121	0.001
$\gamma_{x,a}$	0.183	0.060	0.002
$\gamma_{x,b}$	−1.299	0.499	0.009
$\sigma_{m,a}$	0.005	0.001	0.000
$\sigma_{m,b}$	0.010	0.002	0.000

Notes: Estimated results for the endogenously switching monetary policy rule. SE stands for standard error.

4.5 Table 3: estimated policy-rule parameters

Table 3 takeaway (parameter estimates).

- Persistence: $\gamma_m \approx 0.391$.
- Inflation stabilization: $\gamma_\pi \approx -0.397$ (higher inflation $\Rightarrow$ lower M2 growth).
- Output-gap response is *state dependent*:
  - when $g_x - g_x^* \geq 0$: $\gamma_{x,a} \approx 0.183$,
  - when $g_x - g_x^* < 0$: $\gamma_{x,b} \approx -1.299$.
- Policy shocks are more volatile in the shortfall regime: $\sigma_{m,b} > \sigma_{m,a}$.

Interpretation: the PBoC responds aggressively when growth is below target (because the shortfall term is negative and $\gamma_{x,b}$ is negative, the product becomes positive: M2 growth rises when growth falls short).

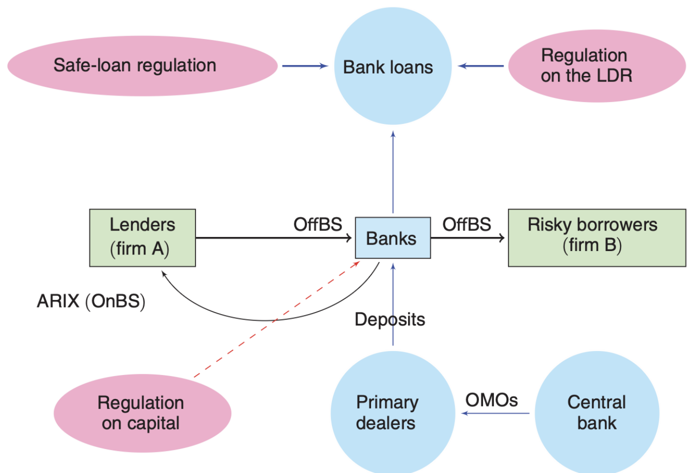

FIGURE 5. AN ILLUSTRATION OF THE NEXUS BETWEEN MONETARY POLICY, THE BANKING SYSTEM, AND SHADOW BANKING

Notes: Banks refers to nonstate banks. Safe-loan regulation refers to a series of 2010–2013 laws that strictly banned commercial banks from expanding bank loans to risky industries. OffBS refers to off balance sheet, OnBS on the balance sheet, and OMOs open market operations. The dashed line arrow indicates that the current capital requirement regulation is far from binding.

5 Micro evidence I: entrusted lending responds to monetary policy

5.1 Data validation: Figure 6

Fig. 6 compares counts of entrusted-loan announcements in the authors' hand-collected data vs. PBoC aggregate statistics. **Interpretation:** the announcement-based micro data capture the time-series variation in entrusted lending reasonably well.

5.2 Baseline trustee regression for total entrusted lending (Eq. (2))

For trustee b and quarter t, the baseline regression is:

$$\log \ell_{b,t} = \alpha + \alpha_g \, g_{t-1} + \beta_{nsb} \, g_{t-1} \, \mathbb{I}(\mathrm{NSB}_b) + \beta_{sb} \, g_{t-1} \, \mathbb{I}(\mathrm{SB}_b) + \mathrm{Controls}_{b,t} + u_{b,t}. \tag{5}$$

Key variables (my notes).

- $\ell_{b,t}$: total volume of entrusted loans intermediated by trustee b in quarter t.
- g_{t-1}: the annual change in the *exogenous* M2 supply in the previous year (constructed from the policy shocks).
- $\mathbb{I}(\mathrm{NSB}_b)$: indicator for *nonstate bank* trustee.
- $\mathbb{I}(\mathrm{SB}_b)$: indicator for *state bank* trustee.

Reading the signs. A *tightening shock* corresponds to lower exogenous M2 growth ($g_{t-1} \downarrow$). Thus:

- if the coefficient on g_{t-1} is *positive*, tightening reduces entrusted lending;
- if the coefficient is *negative*, tightening increases entrusted lending.

5.3 Table 4: results for total entrusted lending

What Table 4 shows.

- **Column (1): all trustees.** Nonbank trustees are the baseline; state-bank and nonstate-bank interactions capture differential responses.
- **Column (2): bank trustees only.** State banks are the baseline; the interaction term captures how nonstate banks differ.
- **Column (3): bank trustees + bank characteristics.** Interactions with bank characteristics probe which constraints amplify the response.

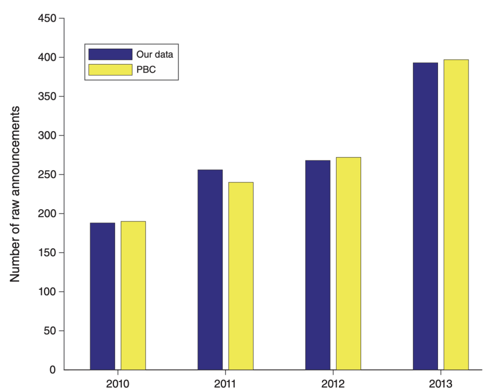

FIGURE 6. CROSS-CHECK ON THE NUMBER OF RAW ANNOUNCEMENTS OF ENTRUST LENDING

Notes: The number of raw announcements we collected is plotted against the number published by the PBC's CFSRs. There were no numbers published by the PBC in years other than 2010–2013.

Sources: PBC and WIND

TABLE 4—ESTIMATED RESULTS FOR PANEL REGRESSIONS ON TOTAL ENTRUSTED LENDING

log of total entrusted lending:	Trustee sample (1)	Bank sample (2)	Bank sample (3)
$g_{t-1} : \alpha_g$	13.12	16.50	253.6
	(7.51)	(10.14)	(85.76)
$g_{t-1}\ \mathcal{J}(NSB_b) : \beta_{nsb}$	−23.77	−30.46	−55.30
	(7.70)	(9.838)	(13.49)
$g_{t-1}\ \mathcal{J}(SB_b) : \beta_{sb}$	4.306		
	(10.72)		
$g_{t-1} \times$ *Liquidity*			0.51
			(0.90)
$g_{t-1} \times$ *ROA*			90.47
			(30.22)
$g_{t-1} \times$ *Size*			−18.87
			(6.35)
$g_{t-1} \times$ *Capital*			−4.17
			(2.59)
$g_{t-1} \times$ *NPL*			9.62
			(14.77)
$g_{t-1} \times$ *LDR*			−0.37
			(0.590)
Observations	583	342	342
Single term controls for bank attributes	No	No	Yes
GDP, deflator controls	Yes	Yes	Yes
Impacts of money growth via state banks	17.42	16.50	37.33
	(9.29)	(10.14)	
Impacts of money growth via nonstate banks	−10.65	−14.00	−17.97
	(4.39)	(5.73)	

Notes: This table shows that nonstate banks facilitated more entrusted lending during monetary policy tightening while state banks did not. The table presents estimates from the panel regressions in which the dependent variable is the logarithm of total entrusted loan amount. Column 1 uses a total of 583 trustee-quarter observations from 2009:I to 2015:IV, including observations of entrusted loans facilitated by nonbank trustees. Columns 2 and 3 report the regression results based on a smaller sample with bank-specific attributes. The smaller sample has a total of 342 bank-quarter observations from 2009:I to 2015:IV. The bank-specific attributes include LDR, size, capital level, liquidity ratio, return on assets, and non-performance loan ratio. We also include GDP at $t-1$ (an annual change of GDP in the previous year) and Inf at $t-1$ (an annual change of the GDP deflator in the previous year) as control variables. A constant term is included but its coefficient is not reported. Standard errors are reported in parentheses.

Main pattern (all specifications).

- The differential response for **nonstate banks** is **negative** and statistically significant: e.g. $\beta_{nsb} \approx -23.8$ (col. 1) or -30.5 (col. 2).
- Translating into total effects (reported in the table):
  - *Impact via state banks:* positive (entrusted lending falls with tightening),
  - *Impact via nonstate banks:* negative (entrusted lending rises with tightening).

Interpretation: tightening pushes entrusted lending activity *toward* nonstate banks (or makes nonstate banks expand their role as trustees). This is consistent with regulatory arbitrage / balance-sheet management rather than a simple "credit demand falls" story.

5.4 Average entrusted lending and the intensive margin (Eq. (3))

To see whether the effect comes from fewer/more deals versus deal size, the paper estimates:

$$S_{b,t} = \alpha + \alpha_g\, g_{t-1} + \beta_{nsb}\, g_{t-1}\, \mathbb{I}(\mathrm{NSB}_b) + \beta_{sb}\, g_{t-1}\, \mathbb{I}(\mathrm{SB}_b) + \mathrm{Controls}_{b,t} + u_{b,t}, \tag{6}$$

where $S_{b,t}$ is either (i) the log average entrusted-loan size or (ii) the number of entrusted loans.

TABLE 5—ESTIMATED RESULTS FOR PANEL REGRESSIONS ON AVERAGE ENTRUSTED LENDING

log of average entrusted lending:	Trustee sample (1)	Bank sample (2)	Bank sample (3)
$g_{t-1} : \alpha_g$	13.24	12.95	175.4
	(6.38)	(8.41)	(79.76)
$g_{t-1}\mathcal{I}(NSB_b) : \beta_{nsb}$	−21.80	−27.63	−34.85
	(6.61)	(8.12)	(16.10)
$g_{t-1}\mathcal{I}(SB_b) : \beta_{sb}$	3.36		
	(9.26)		
$g_{t-1} \times$ *Liquidity*			−0.32
			(0.82)
$g_{t-1} \times$ *ROA*			87.38
			(27.25)
$g_{t-1} \times$ *Size*			−12.98
			(6.12)
$g_{t-1} \times$ *Capital*			−6.88
			(3.10)
$g_{t-1} \times$ *NPL*			12.91
			(12.42)
$g_{t-1} \times$ *LDR*			−0.25
			(0.53)
Observations	583	342	342
Single term controls for bank attributes	No	No	Yes
GDP, deflator controls	Yes	Yes	Yes
Impacts of money growth via state banks	16.60	12.95	17.75
	(7.96)	(8.41)	
Impacts of money growth via nonstate banks	−8.56	−14.68	−17.09
	(4.18)	(4.09)	

Notes: This table shows that nonstate banks facilitated more entrusted lending on average (intensive margin) during monetary policy tightening while state banks did not. The table presents estimates from the panel regressions in which the dependent variable is the logarithm of average entrusted loan amount. Column 1 uses a total of 583 trustee-quarter observations from 2009:I to 2015:IV, including observations of entrusted loans facilitated by non-bank trustees. Columns 2 and 3 report regression results based on a smaller sample with bank-specific attributes. The smaller sample has a total of 342 bank-quarter observations from 2009:I to 2015:IV. The bank-specific attributes include LDR, size, capital level, liquidity ratio, return on assets, and non-performance loan ratio. We also include GDP at $t-1$ (an annual change in GDP in the previous year) and Inf at $t-1$ (an annual change in the GDP deflator in the previous year) as control variables. A constant term is included but its coefficient is not reported.

TABLE 6—SUMMARY STATISTICS OF BANK-SPECIFIC ATTRIBUTES FROM 2009 TO 2015

Attribute/variable	Observations	Mean	SD	Min	Median	Max
LDR	396	69.49	7.11	47.43	71.70	85.16
Size	396	14.86	1.20	11.42	14.84	16.84
Capital	396	6.05	1.25	3.18	5.99	12.34
Liquidity	396	27.01	7.13	12.21	25.44	48.10
ROA	396	1.06	0.19	0.42	1.06	1.58
NPL	396	1.08	0.56	0.38	0.96	4.32

Note: All variables except size (log value of total assets) are expressed in percent.

Source: Bankscope

5.5 Table 5: results for average entrusted loan size

Table 5 takeaway.

- The same sign pattern shows up on the *intensive margin*: for nonstate banks the total effect is negative (loan size rises when M2 tightens).
- Therefore, tightening appears to increase entrusted lending by nonstate banks partly through **larger average loans** (not only counts).

5.6 Bank heterogeneity (Eq. (4)) and Table 6

When restricting to banks, the regression allows interactions with bank characteristics:

$$\log \ell_{b,t} = \alpha + \alpha_g\, g_{t-1} + \beta_{nsb}\, g_{t-1}\, \mathbb{I}(\mathrm{NSB}_b) + \left(g_{t-1} \cdot \mathbf{Z}_{b,t-1}\right)' \boldsymbol{\delta} + \mathrm{Controls}_{b,t} + u_{b,t}. \tag{7}$$

Table 6 provides summary statistics for $\mathbf{Z}_{b,t}$ (size, capital, liquidity, ROA, NPL, LDR, etc.).
Interpretation.

- The nonstate-bank interaction remains strongly negative even after conditioning on bank balance-sheet characteristics.

- Some characteristics (e.g. size/capital) help explain which banks respond more, consistent with constraints interacting with monetary policy.

6 Micro evidence II: tightening increases on-balance-sheet shadow banking (ARIX)

6.1 ARIX regression (Eq. (5))

Let $\xi_{b,t}$ denote bank b's ARIX holdings. The regression is:

$$\log \xi_{b,t} = \alpha + \alpha_g \, g_{t-1} + \beta_{nsb} \, g_{t-1} \, \mathbb{I}(\mathrm{NSB}_b) + \big(g_{t-1} \cdot \mathbf{Z}_{b,t-1}\big)' \boldsymbol{\delta} + \mathrm{Controls}_{b,t} + \varepsilon_{b,t}. \tag{8}$$

6.2 Table 7: ARIX results

Main pattern in Table 7.

- The nonstate-bank interaction is large and negative (e.g. around -64), implying:

$$\text{Effect for nonstate banks} = \alpha_g + \beta_{nsb} < 0.$$

- In other words, when exogenous M2 growth falls, **nonstate banks increase ARIX** (more on-balance-sheet shadow assets), while state banks move in the opposite direction.

Interpretation: tightening does not just shift *off*-balance-sheet activity; it also increases *on*-balance-sheet shadow-banking assets at nonstate banks.

Table 7—Estimated Results for Panel Regressions on ARIX

log of ARIX:	Bank sample (1)	Bank sample (2)	Bank sample (3)	Bank sample (4)
$g_{t-1} : \alpha_g$	26.56	23.57	245.9	341.1
	(12.55)	(13.01)	(189.8)	(215.3)
$g_{t-1}\mathcal{I}(NSB_b) : \beta_{nsb}$	−64.26	−69.30	−108.6	−113.8
	(15.45)	(15.48)	(32.83)	(32.59)
$g_{t-1} \times$ *Liquidity*			1.006	0.992
			(1.15)	(1.35)
$g_{t-1} \times$ *ROA*			196.2	190.7
			(93.56)	(87.09)
$g_{t-1} \times$ *Size*			−24.26	−27.34
			(16.60)	(18.40)
$g_{t-1} \times$ *Capital*			−13.29	−14.73
			(12.47)	(12.89)
$g_{t-1} \times$ *NPL*			−18.74	−29.19
			(24.62)	(21.09)
$g_{t-1} \times$ *LDR*			0.731	0.448
			(1.37)	(1.29)
Observations	410	373	373	373
Single term controls for bank attributes	No	No	Yes	Yes
GDP, deflator controls	Yes	Yes	Yes	Yes
Impacts of money growth via state banks	26.56	23.57	66.27	70.12
	(12.55)	(13.01)		
Impacts of money growth via nonstate banks	−37.69	−45.73	−42.33	−43.68
	(9.85)	(8.89)		

Notes: This table shows that nonstate banks invested in more ARIX on their balance sheets during monetary policy tightening while state banks did not. The table presents estimates from the panel regressions in which the dependent variable is ARIX, which includes all shadow banking products on the balance sheet. The sample period is from 2009:I to 2015:IV. Column 1 uses a total of 410 bank-quarter observations from 2009:I to 2015:IV without inclusion of bank-specific attributes. Columns 2–4 report regression results based on a smaller sample with bank-specific attributes. The smaller sample has a total of 373 bank-quarter observations from 2009:I to 2015:IV. Column 3 reports the regression results with the end-of-year LDR. Column 4 reports the regression results with the average of daily LDR. We also include GDP at $t-1$ (an annual change in GDP in the previous year) and Inf at $t-1$ (an annual change in the GDP deflator in the previous year) as control variables. A constant term is included but its coefficient is not reported. Standard errors are reported in parentheses.

Table 8—Correlation between New Entrusted Loans ($\mathcal{L}$) Channeled by Banks and Changes in ARIX or the Share of ARIX in 2009–2015

Description	State banks	p-value	Nonstate banks	p-value
$\mathrm{corr}(\Delta\mathrm{ARIX}, \mathcal{L})$	0.224	0.197	0.621	0.000
$\mathrm{corr}\left(\frac{\mathrm{ARIX}}{\mathrm{ARIX}+B}, \mathcal{L}\right)$	−0.179	0.304	0.458	0.001

Note: B stands for bank loans.

Source: Our constructed microdata at the level of individual banks

6.3 Table 8: connecting off- and on-balance-sheet shadow banking

Table 8 reports correlations between (i) new entrusted loans facilitated by banks and (ii) changes in ARIX.

- For **nonstate banks**, correlations are large and significant (e.g. $\text{corr}(\Delta\text{ARIX}, L) \approx 0.62$).
- For **state banks**, correlations are small and statistically insignificant.

Interpretation: nonstate banks appear to actively link the off-balance-sheet entrusted-loan business with on-balance-sheet ARIX accumulation (e.g., by purchasing beneficiary rights or warehousing exposures).

6.4 Figure 7 and Table 9: ARIX shares by bank type

Fig. 7: ARIX share in total bank credit rises dramatically for nonstate banks after 2011, reaching about 29% by 2015, while staying near 2–4% for state banks.

Table 9: yearly distributional summaries confirm this divergence:

- State-bank ARIX shares remain low and stable.
- Nonstate-bank ARIX shares increase sharply and become widely dispersed across banks.

Interpretation: shadow-banking activity (as measured on bank balance sheets) is primarily a *nonstate-bank phenomenon* in this period.

7 Mechanism: a bank model with deposit withdrawals and regulatory constraints

7.1 Linking policy shocks to deposit withdrawals (Eq. (6) + Fig. 8)

The theoretical mechanism uses a simple mapping from monetary policy shocks to the mean deposit-withdrawal shock:

$$\mu(\varepsilon_{m,t}) \simeq -(2\varepsilon_{m,t} + 1). \tag{9}$$

Interpretation: tighter policy (a negative shock in exogenous M2 growth) raises expected deposit withdrawals / reduces deposit growth. This is consistent with **Fig. 8**, which shows bank deposit growth tracking M2 growth closely in the data.

7.2 Within-period timing: lending stage vs. balancing stage

The model has two stages:

- **Lending stage:** the bank chooses its balance sheet (deposits, cash, loans, risky nonloan assets) subject to capital/liquidity constraints.
- **Balancing stage:** after a deposit-withdrawal shock ω (linked to policy) and a default shock ξ on risky nonloan assets, the bank must satisfy the loan-to-deposit ratio (LDR) regulation.

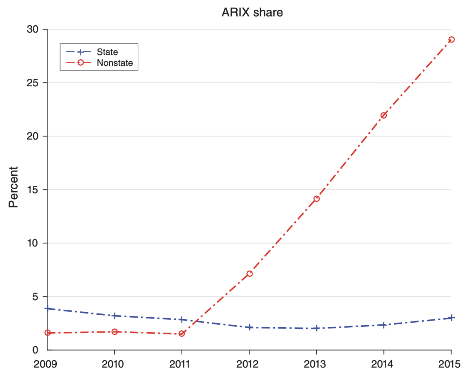

FIGURE 7. SHARE OF ARIX IN TOTAL BANK CREDIT

Notes: ARIX share is the ratio of ARIX holdings to the sum of ARIX holdings and bank loans on the balance sheets of the 16 publicly listed commercial banks. Based on the bank asset data from these individual banks, the data are categorized into state banks as one group and nonstate banks as the other group. ARIX holdings and bank loans are aggregate values for each group.

Sources: PBC and WIND

TABLE 9—ARIX SHARES IN 2009–2015 (*Percent*)

	State banks				Nonstate banks			
Year	p30	Median	Average	p70	p30	Median	Average	p70
2009	2.82	3.89	3.87	4.79	0.50	1.68	1.59	4.56
2010	1.77	3.35	3.20	4.43	0.65	1.12	1.71	6.78
2011	1.03	3.02	2.84	3.54	0.59	1.13	1.50	2.20
2012	1.78	2.00	2.10	2.27	3.35	9.25	7.14	11.37
2013	1.64	1.86	2.02	2.58	9.84	17.67	14.18	18.70
2014	1.29	1.86	2.35	4.12	17.98	20.80	21.98	25.88
2015	2.18	2.27	2.98	5.31	20.58	22.28	29.03	31.06

Note: The (weighted) average values are computed in the same way as in Figure 7. The distribution demarcations p30 and p70 stand for the thirtieth and seventieth percentiles. These percentiles, as recommended by Sims and Zha (1999), contain two thirds of the pointwise probability for the distribution of each group of banks.

Source: Data source is the same as in Figure 7.

7.3 Balancing stage: LDR violation variable and law of motion (Eqs. (7)–(10))

Let q be the price of bank loans B (so qB is the market value of loans). The LDR regulation is:

$$qB \leq \theta(1-\omega)\frac{D}{R^D}, \tag{10}$$

and define the violation amount:

$$x = qB - \theta(1-\omega)\frac{D}{R^D}. \tag{7}$$

If $x > 0$, the bank violates the LDR and must raise deposits (at a cost) or shrink exposures. The post-shock state evolves as:

$$\tilde{D}' = D(1-\omega) + \chi(x) - \xi R^D I_r, \tag{8}$$

$$\tilde{C}' = C - \omega D, \tag{9}$$

$$\tilde{B}' = B. \tag{10}$$

Interpretation.

- $\chi(x)$ captures costly deposit recouping (how much deposits the bank can raise to fix an LDR violation).
- A tighter policy shock makes ω larger on average, pushing x upward and making the LDR more likely to bind.
- When the LDR binds, the bank wants to reduce measured loans B *or* shift activity into assets not counted as loans (shadow-banking assets).

7.4 Lending stage: bank problem (Eqs. (11)–(16)) and balance sheet (Eq. (17))

The lending-stage value function is:

$$V^\ell(\tilde{C}, \tilde{B}, \tilde{D}; \varepsilon_m) = \max\ U(DIV) + \mathbb{E}_{\omega,\xi}\Big[V^b(C, B, D; \varepsilon_m)\Big]. \tag{11}$$

The key constraints/definitions include:

$$\frac{D}{R^D} - \tilde{D} + (1 - q_r)I_r + (1 - q)S - DIV = \phi + I_r + (B - \tilde{B}), \tag{12}$$

$$\frac{D}{R^D} \leq \kappa\left(C + q_r I_r + qB - \frac{D}{R^D}\right), \tag{13}$$

$$C \geq \psi\left(C + q_r I_r + qB - \frac{D}{R^D}\right), \tag{14}$$

$$C = \tilde{C} + \phi, \tag{15}$$

$$B = \delta\tilde{B} + S. \tag{16}$$

Balance sheet identity at the lending stage:

$$\frac{D}{R^D} + e - DIV = C + q_r I_r + qB. \tag{17}$$

High-level mechanism from the model (my summary).

- A monetary tightening shock reduces deposit funding (via the induced withdrawal shock).
- With LDR/capital/liquidity constraints, constrained banks (nonstate banks) substitute away from measured loans B and toward shadow-banking exposures I_r (captured empirically by entrusted lending and ARIX).
- The substitution weakens the aggregate credit contraction from monetary policy.

8 Aggregate implication: monetary policy shocks and total credit (Eq. (18) + Fig. 9)

8.1 Dynamic panel VAR (Eq. (18))

The paper estimates a bank-level dynamic panel VAR of the form:

$$Y_t = \Phi_{y,1}Y_{t-1} + \cdots + \Phi_{y,p}Y_{t-p} + \Phi_{m,0}\varepsilon_{m,t} + \cdots + \Phi_{m,q}\varepsilon_{m,t-q} + e_t, \tag{18}$$

where Y_t includes outcomes like bank-loan growth and ARIX growth, and $\varepsilon_{m,t}$ is the monetary policy shock from the M2 rule.

8.2 Figure 9: impulse responses

Fig. 9 reports responses to a *one-standard-deviation fall* in exogenous M2 growth (tightening):

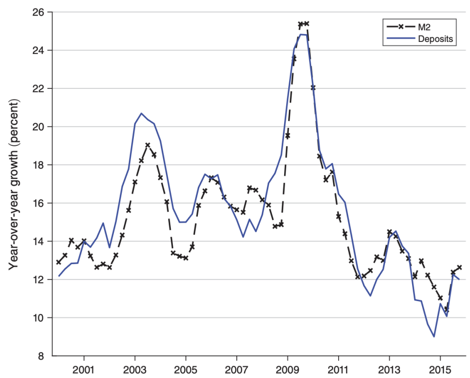

Figure 8. Quarterly Time Series of M2 Growth (*Year-over-Year*) and Growth of Bank Deposits (*Year-over-Year*)

Sources: PBC and CEIC

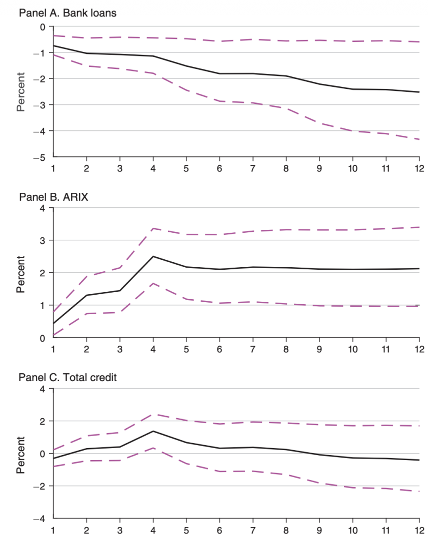

Figure 9. Estimated Quarterly Dynamic Responses to a One-Standard-Deviation Fall of Exogenous M2 Growth

Notes: The estimation is based on our constructed banking data. Solid lines are the estimates and dashed lines represent the 0.68 probability bands as advocated by Sims and Zha (1999).

- **Bank loans fall** (Panel A): the standard credit-tightening effect.
- **ARIX rises** (Panel B): banks expand on-balance-sheet shadow-banking exposures.
- **Total credit (bank loans + ARIX) does not fall much** and can rise temporarily (Panel C): the shadow-banking response offsets the bank-loan contraction.

Interpretation: the paper's central message in one picture: monetary policy affects the *composition* of credit (toward shadow banking) and therefore the net effect on total credit is muted.

9 Short summary

- Monetary policy shocks in China transmit strongly to **bank deposits** and **traditional bank loans**.
- But banks, especially **nonstate banks**, endogenously substitute toward **shadow banking** (entrusted loans and ARIX) when tightening binds regulatory constraints.
- Therefore, monetary policy has a major **composition effect** in credit markets, and the **total credit effect** can be small.
- Empirically, this is supported by micro regressions (Tables 4–7), cross-measure correlations (Table 8), and impulse responses (Fig. 9).
- Broad takeaway: to understand monetary transmission in modern economies, I need to track not only bank lending but also shadow-banking channels that can expand endogenously when policy tightens.

10 Conclusion

10.1 Summary

Chen, Ren, and Zha (2018) show that, in China's post-2008 environment, **contractionary monetary policy can reduce traditional bank lending while simultaneously triggering a rapid expansion of shadow-banking credit**, thereby **weakening the effectiveness of monetary tightening on total bank credit**. The central empirical claim is that monetary tightening over 2009–2015 contributed to (i) a fall in conventional bank loans and (ii) a rise in shadow-banking lending both *off balance sheet* (entrusted loans) and *on balance sheet* (riskier non-loan assets recorded as ARIX), with the **offsetting expansion primarily driven by nonstate banks**. To identify monetary policy innovations in an institutional setting where China targets M2 growth, the paper develops and estimates an **endogenously switching money-growth rule** and constructs an **exogenous M2 shock series** (from Eq. (1) in these notes). Combining these policy shocks with newly assembled micro banking datasets, the authors provide evidence that the shadow-banking response is not merely a passive reflection of borrower demand, but instead is consistent with **banks actively reallocating credit creation across categories to circumvent binding regulations under tighter funding conditions**.

10.2 Key empirical conclusions

1. **Tightening raises shadow banking while reducing traditional loans.** Contractionary M2 shocks are associated with *higher* shadow-banking activity and *lower* bank loans, implying a strong *composition effect* of monetary policy on bank asset portfolios.

2. **Nonstate banks are the marginal adjusters.** The response of entrusted-lending facilitation and ARIX accumulation is concentrated among **nonstate banks** (Tables 4–7 in these notes), whereas state banks react much less in the same directions.

3. **Total credit becomes less sensitive to monetary tightening.** In dynamic responses (panel VAR; Eq. (18)), the increase in ARIX can offset the decline in bank loans, so that **total bank credit does not fall much** (Figure 9 in these notes).

4. **Off–on balance-sheet linkage matters.** The strong correlation between entrusted lending and ARIX growth for nonstate banks (Table 8) supports the view that banks are not only intermediating shadow credit off balance sheet; they are also **bringing shadow exposures onto balance sheet** in response to tightening.

10.3 Mechanism intuition: why tightening can expand shadow banking

The mechanism is a regulatory-arbitrage response to a funding squeeze:

1. **Monetary tightening reduces deposit funding (liability-side contraction).** With a quantity-based M2 regime, a negative exogenous M2 shock is mapped into tighter liquidity and weaker deposit growth (consistent with Fig. 8 in these notes).

2. **Binding constraints apply disproportionately to "standard loans."** Traditional bank loans are directly subject to regulatory constraints such as the loan-to-deposit ratio (LDR) and "safe-loan" restrictions (modeled in the lending/balancing stages and the LDR-violation object x in Eq. (7)).

3. **Banks substitute toward less-constrained credit categories.** When deposits tighten and the LDR becomes more likely to bind, banks (especially nonstate banks) have incentives to:

   - *facilitate more off-balance-sheet shadow lending* (entrusted loans), and
   - *increase on-balance-sheet non-loan risky assets* recorded as ARIX, which can bypass some loan-specific regulations.

4. **Institutional asymmetry explains heterogeneous bank responses.** The paper argues that the strong nonstate-bank response is consistent with differential constraints/incentives and an uneven ability/willingness to engage in regulatory circumvention (e.g., profit motives, bindingness of deposit constraints, and institutional mandates).

Bottom line intuition: in this environment, monetary policy does not only "shift the level of credit"; it also **reallocates credit creation across categories**, and shadow banking becomes an internal margin of adjustment that can blunt the aggregate tightening.

10.4 Policy implications

The conclusion emphasizes that the old credit-control architecture (a quantity-based regime working through the bank lending channel) was designed *before* shadow banking became large. Once banks can expand shadow-banking exposures (especially via ARIX), tightening M2 is less effective at restraining *total* bank credit. The paper highlights three policy directions:

1. **Macroprudential coordination:** regulating *broad credit* growth (not just loans) so that it does not systematically deviate from the M2 target is "moving in the right direction" (as discussed immediately before Section VI in the paper).

2. **Modernizing the monetary framework:** moving toward a policy interest rate as an intermediate target, rather than relying primarily on quantity-based M2 control.

3. **Regulatory redesign:** removing the LDR regulation and strengthening capital adequacy requirements with *risk-appropriate* treatment of ARIX and related shadow assets—noting that this is difficult because ARIX composition and implicit guarantees are opaque.

10.5 Limitations and what the paper flags for future research

The paper explicitly notes that it focuses on the **banking sector balance sheet** (asset-side reallocation) and abstracts from other important dimensions. Two key gaps are:

- **Real-economy transmission:** whether the bank lending channel propagates to the real economy differently through bank loans versus ARIX/shadow exposures remains an open question.

- **Broader reform interactions:** financial liberalization can expand "financing flexibility" and erode the effectiveness of well-intended policies; the authors point to the need for a macroprudential framework that coordinates with monetary policy as liberalization proceeds.

10.6 Takeaway

In China's quantity-based monetary regime, tightening M2 reduces traditional bank loans but induces nonstate banks to expand shadow-banking credit (off-balance-sheet entrusted lending and on-balance-sheet ARIX), so the main effect of monetary policy is a powerful ***recomposition*** *of bank credit that can blunt the intended contraction in total bank credit.*